

About Simulation Execution Standard - (SES)

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Architecture: SIMULOS® — Simulation Governance Architecture Category:
Governance & Enforcement Subcategory: Simulation Governance Parent Standard:
Simulation Execution Standard (SES) Governed Space: Simulation Execution
Operational Space: Simulation Execution Governance Governance Flow Position: 8
of 12 — Simulation Scenario → Simulation Execution → Simulation Output
Governance

Canonical Definition

Simulation Execution Standard (SES) defines the governance conditions under which a governed simulation is instantiated, executed, controlled, observed, repeated where required, and documented so that the actual execution remains traceable to the Simulation Object, model, environment, scenario, inputs, parameters, configuration, runtime conditions, interventions, deviations, and outputs that formed the execution event.

SES governs the point at which a designed simulation becomes an actual simulation execution.

It ensures that organizations can distinguish between:

what was intended to run

and:

what actually ran.

Why SES Matters

Modern organizations may execute thousands or millions of simulations across AI systems, autonomous systems, robotics, digital twins, engineering environments, and mission systems.

The challenge is therefore not only whether simulation capability exists.

The governance question becomes:

Can the organization reconstruct exactly what happened during a materially relevant simulation execution?

SES provides the structure for answering:

Which configuration actually ran?

Which model and environment versions were used?

Which scenario, inputs, and parameters were instantiated?

What happened during execution?

Were there interventions, interruptions, failures, or deviations?

Was the simulation repeated under comparable conditions?

Can the resulting outputs be traced to the execution that produced them?

What SES Governs

SES governs five connected execution spaces:

Simulation Instantiation Establishes the exact executable simulation instance and binds it to its relevant configuration.

Simulation Execution Control Governs how execution is initiated, progressed, controlled, interrupted, modified where permitted, terminated, and completed.

Simulation Observation Preserves materially relevant states, events, transitions, anomalies, interventions, and runtime conditions.

Simulation Repetition Governs repeated runs, randomization, repeatability, reproducibility conditions, execution variance, and comparability.

Simulation Execution Documentation Creates the traceable record needed to reconstruct what actually occurred.

Governance Flow

Governed Scenario & Simulation Configuration

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Simulation Instantiation

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Execution Control

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Simulation Observation

↓

Simulation Repetition

↓

Execution Documentation

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Governed Simulation Execution Record



Simulation Output Governance What SES Does Not Determine

SES does not determine whether: the Simulation Object was correctly selected, the representation perfectly reflects reality, the simulation assumptions are universally valid, the simulation output is correct, an inference from the output is justified, the simulated system is safe, or the system should be deployed.

Those questions belong to other SIMULOS® spaces or compatible OOF® architectures.

SES governs specifically:

the execution event and its reconstructability.

Enterprise Application

SES can operate above an organization's existing simulation infrastructure.

A company does not need to replace its: simulation engine, robotics simulator, digital twin, Monte Carlo environment, HPC infrastructure, AI evaluation environment, engineering simulator, or proprietary simulation platform.

SES provides a governance layer for examining whether simulation execution remains identifiable, controlled, observable, repeatable where required, documented, and traceable.

This makes SES applicable both as an execution governance framework and as a diagnostic structure for identifying execution-governance gaps in existing simulation environments. Use Case 1 — Autonomous System SimulationScenario

A company performs thousands of simulations before deploying an autonomous robotic system.

Several runs show unexpected collision-avoidance behavior, but engineers cannot reliably determine whether the difference originated from: model version, environment configuration, sensor inputs, randomization, parameter changes, runtime intervention, or actual autonomous behavior.

SES Application

SES binds each material execution to its:

Execution ID → Model → Environment → Scenario → Inputs → Parameters → Runtime Conditions → Observations → Outputs

Repeated runs remain distinguishable, while material execution deviations and interventions remain visible. Result

The organization can reconstruct the simulation conditions associated with the anomalous behavior and determine which executions are legitimately comparable before downstream interpretation or validation begins. Use Case 2 — Space Mission Simulation CampaignScenario

A space organization executes a large mission-simulation campaign covering combinations of: navigation uncertainty, communication loss, sensor degradation, power limitations, subsystem failures, autonomous recovery, and changing environmental conditions.

A critical mission decision depends upon results generated across a very large population of simulation runs.

SES Application

SES establishes execution and campaign traceability so that materially relevant results remain connected to:

Scenario → Configuration → Execution Conditions → Runtime Events → Repetition
→ Completion Status → Output

Interrupted, failed, modified, or anomalous runs remain visible rather than disappearing from the analytical record. Result

Decision-makers can determine not merely that simulations were performed, but which configurations actually executed, under which conditions, whether execution remained complete and controlled, and which exact executions produced the outputs being relied upon.

Business Value

SES can help organizations identify potential governance gaps such as:

Execution Traceability Gaps

Configuration-to-Run Gaps

Observation Gaps

Intervention Traceability Gaps

Execution Drift

Repetition Comparability Gaps

Failed-Run Visibility Gaps

Execution-to-Output Lineage Gaps

These findings can then be investigated before conclusions are drawn about simulation quality, validity, safety, or operational readiness.

Relationship to Adjacent SIMULOS® Standards

Simulation Scenario Standard (SSS) asks:

What scenarios must be simulated?

Simulation Execution Standard (SES) asks:

What actually happened when those simulations were executed?

Simulation Output Governance Standard (SOGS) then asks:

What exactly did those executions produce, and what may legitimately be inferred from those outputs?

The architectural transition is therefore:

Scenario → Execution → Output

Foundational Principle

A simulation result cannot be fully understood from its output alone; it must remain connected to the execution conditions and execution history that produced it.

Canonical Closing Statement

Simulation Execution Standard (SES) provides the governance structure through which simulation intent becomes traceable simulation activity. By governing instantiation, execution control, observation, repetition, and execution documentation, SES enables organizations to reconstruct what actually ran, distinguish intended configuration from executed configuration, preserve material deviations and failures, compare repeated executions appropriately, and maintain lineage between simulation executions and the outputs they produce.

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